OMNI CAMERA ROVER | REV B PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

File: interlock.kicad_sch

Battery protection

File: battery_protection.kicad_sch

PWR_FLAG GND

PWR_FLAG VMOTOR

PWR_FLAG USB_5V

PWR_FLAG U2_VINT

PWR_FLAG U3_VINT

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

Nathaniel Sanford

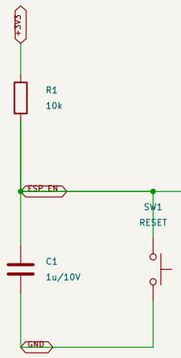
Sheet: /
File: omni_ugv.kicad_sch

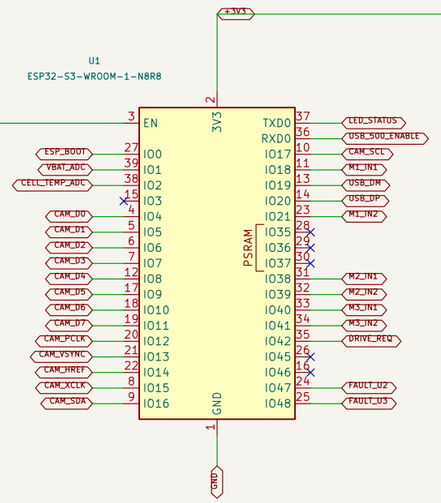
Title: Omni UGV / Rev B

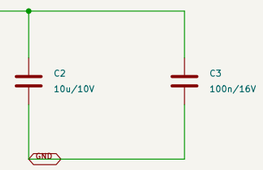
Size: A3	Date: 2026-11-02	**Rev: B-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 1/7

Controller, camera and service points

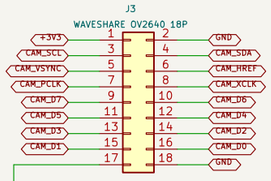
RESET / enable RC







Waveshare OV2640 | documented 2 x 9 interface



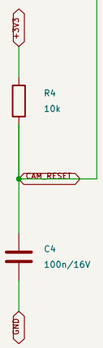
Status indicator



BOOT strap



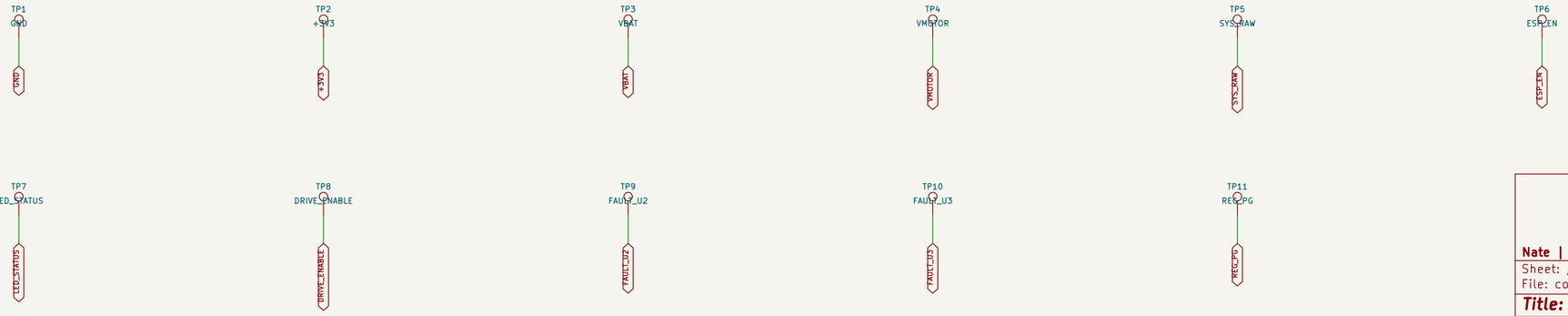
Camera reset RC





[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Service points (named nets continue on other sheets)



Nate | Omni camera rover

Sheet: /1. Controller/
File: controller.kicad_sch

Title: Controller, camera and service points

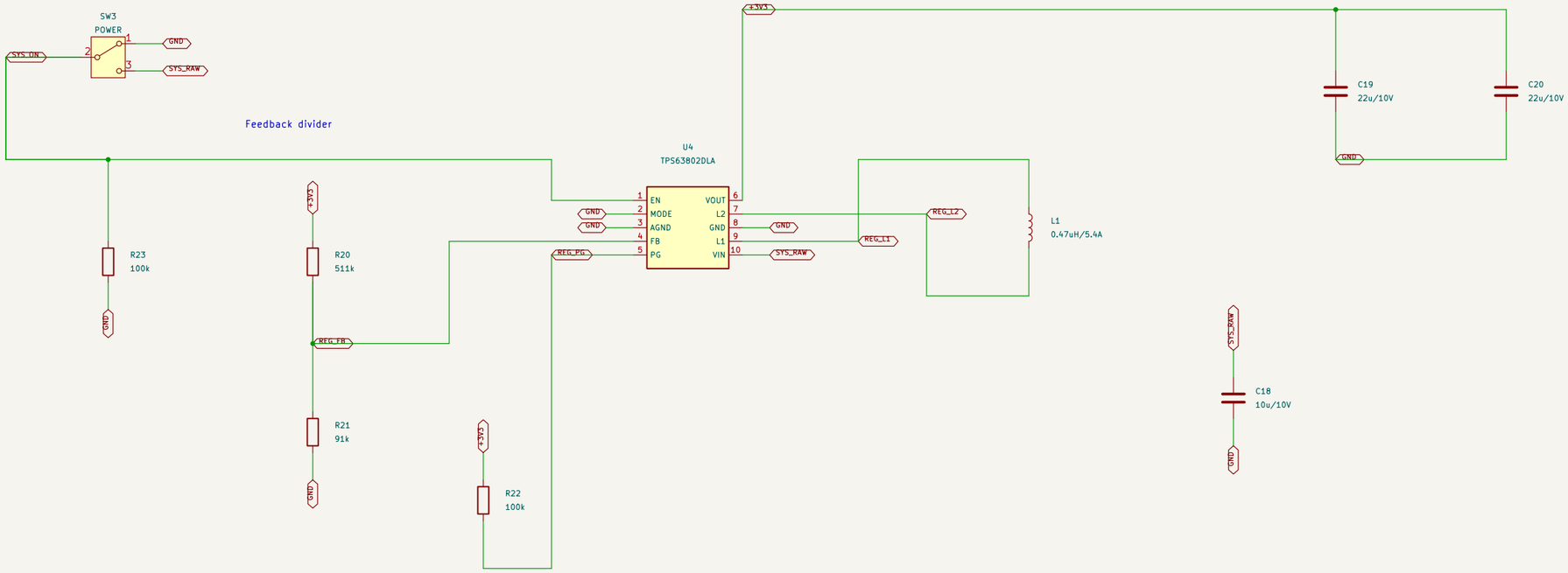
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 2/7

3.3 V buck–boost supply

System enable

Feedback divider

Output capacitors

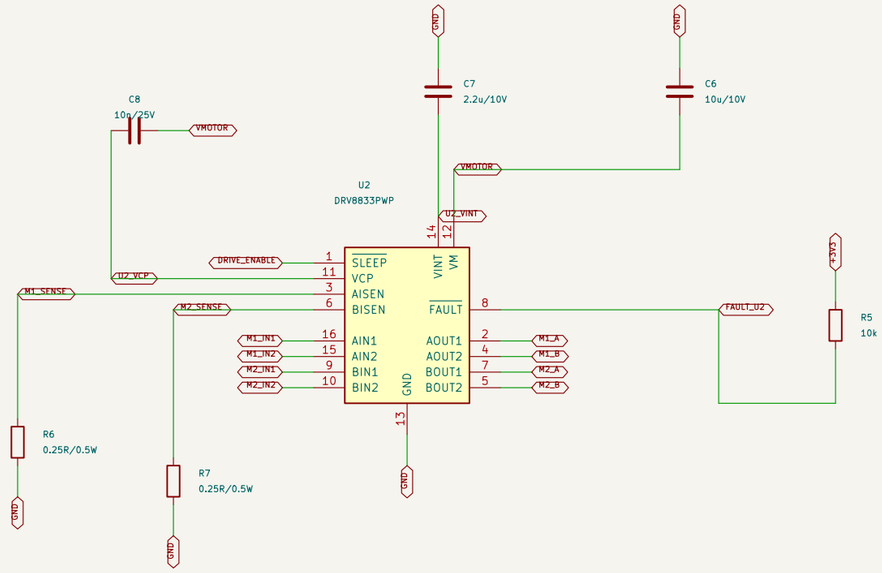


[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

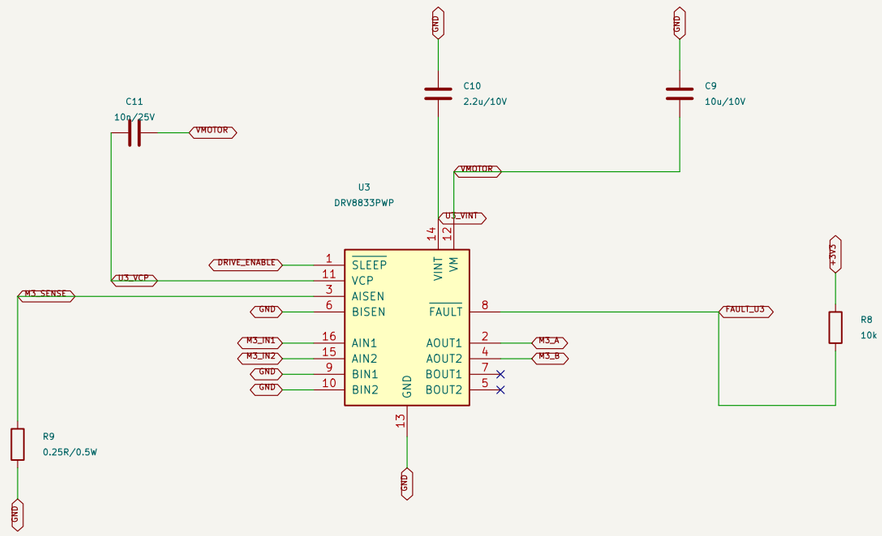
Nate | Omni camera rover		
Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: 3.3 V buck–boost supply		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 3/7

Three independent motor channels

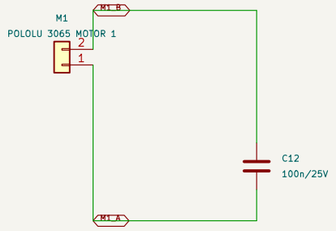
U2: wheels 1 and 2

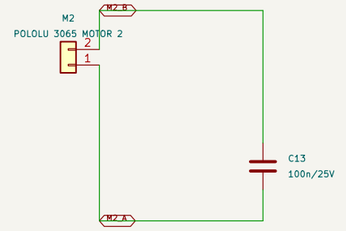


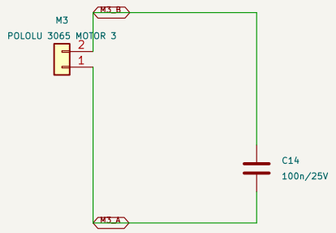
U3: wheel 3; unused bridge pins retained as designed

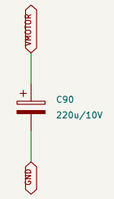


Motor terminal pads and suppression capacitors







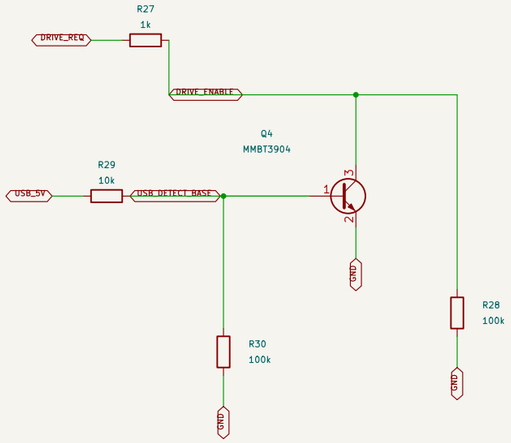


[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

Nate | Omni camera rover		
Sheet: /2. Motors/		
File: motors.kicad_sch		
Title: Three independent motor channels		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 4/7

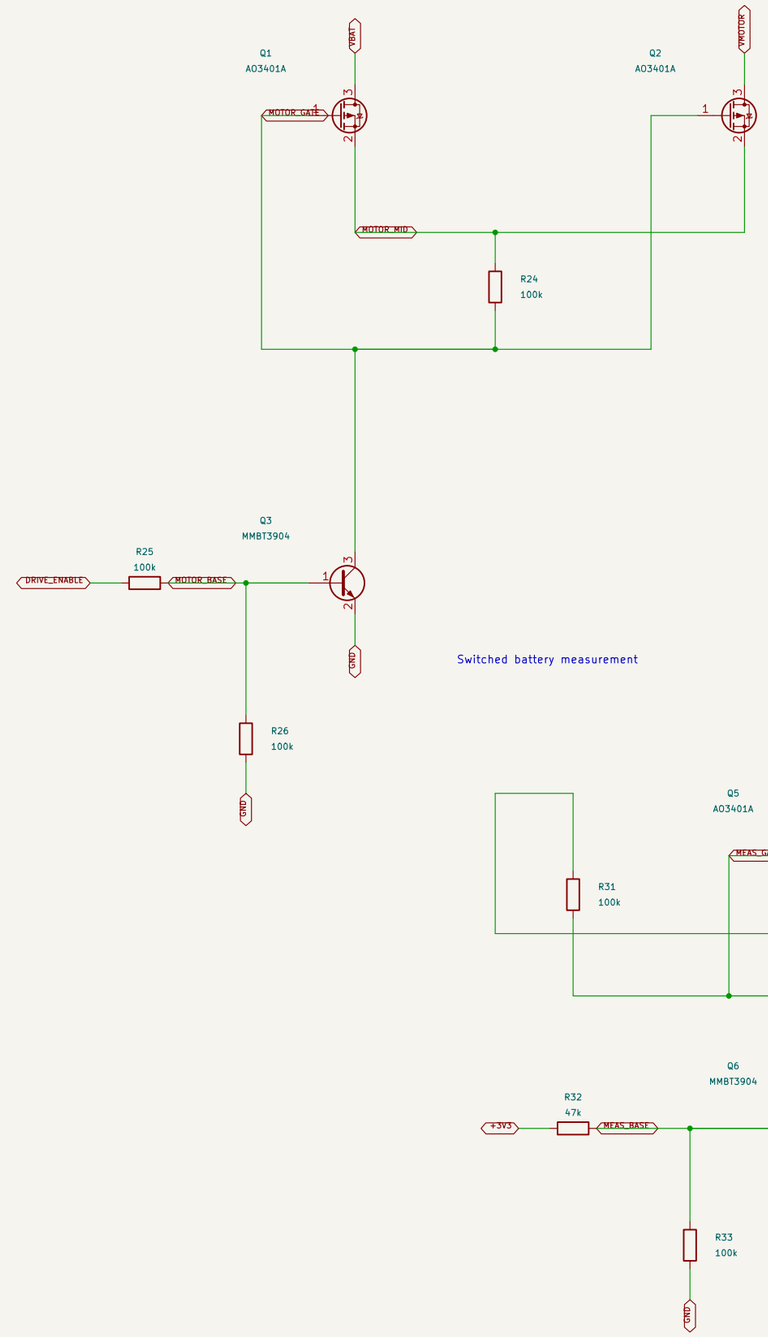
USB motor interlock and battery measurement

USB presence clamps DRIVE_ENABLE

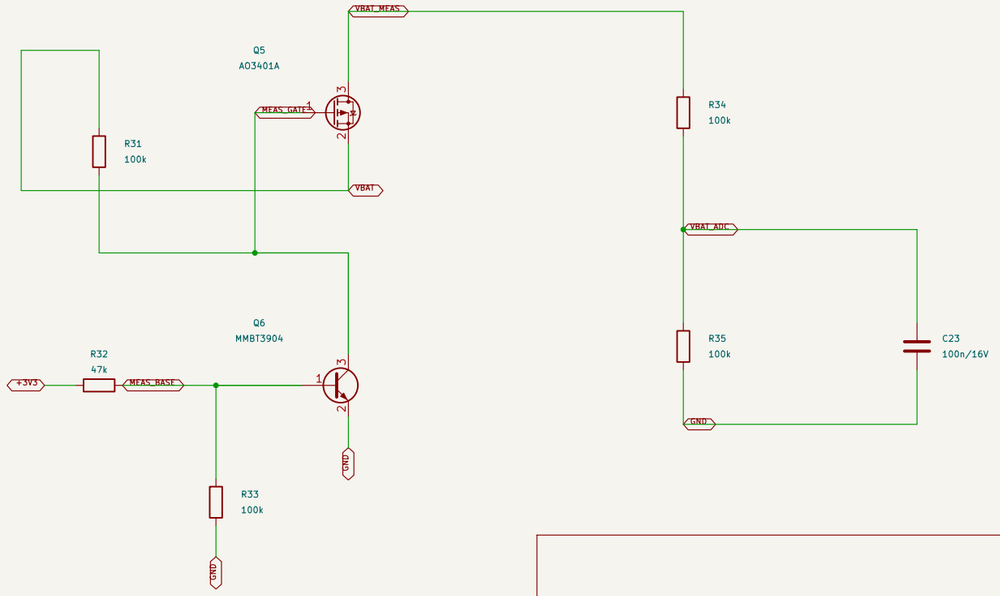


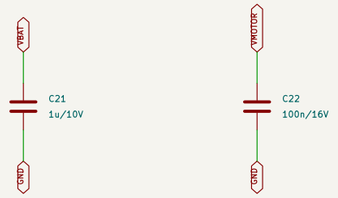
[TARGET] Reset / absent commands keep motors disabled.
[TBD–MEASURE] Verify USB insertion/removal and motor braking transients.

Back-to-back motor supply switch



Switched battery measurement





Nate | Omni camera rover	
Sheet: /5. Interlock/	
File: interlock.kicad_sch	
Title: USB motor interlock and battery measurement	
Size: A3 | Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8	Id: 5/7

USB input, charging and cell connections

USB data to ESP32 native USB

Raw cell –> protection sheet

Charge / system power path

Cell–bonded charger probe

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two–wire 103AT–2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheet: /3. Charging/	
File: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: B – prototype
	Id: 6/7

1S protection and cell-temperature monitoring

Cell voltage / current sensing

R37
100R 1%

C24
100n/16V X7R

R38
20R 1%

U7
BQ298217RUGR

R36
10m 1% 1W

R39
1k 1%

Common-drain high-side protection FET pair

R41
10M

Q8
CSD17308Q3

Q7
CSD17308Q3

R40
10M

TH2 -> J5 -> GPIO1 / ADC1

R42
10k 1%

J5
CELL NTC - DRIVE

R43
1k 1%

TH2
103AT-2 CELL-MOUNTED

C25
100n/16V X7R

TH1 -> J4 charger input

TH1
103AT-2 CELL-MOUNTED

#FLG20
PROTECTION_PWR_FLAG

#FLG21
PROTECTION_PWR_FLAG

[SPEC] TH1 and TH2 are OFF-BOARD, bonded to the cell through electrical insulation.
Separate two-wire probes; do not connect either sensor to raw CELL_N.
[TARGET] Firmware must stop driving on overtemperature, open/short probe or stale readings.

Nate | Omni camera rover	
Sheet: /Battery protection/	
File: battery_protection.kicad_sch	
Title: 1S protection and cell-temperature monitoring	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: B — prototype
	Id: 7/7